

# The $\gamma$ -barrier

## Residuals, projections, and bounds on scale drift

Theory-first collaborator handoff. Expanded derivations; intentionally brief experiments.

**Research question.** Why does training fail to acquire larger, task-relevant inverse widths when those scales would improve approximation? The goal is to understand and improve learned functions, not to reproduce a particular QI parameter pattern.

## 1 The mechanism to distinguish, and the status of the claims

The working picture has three stages. The current geometry has some directions that barely change the fitted function. Readout adjustment may reproduce much of the change that remains. Gradient descent then pairs the residual with the surviving tangent directions; locally, it pays the *square* of their singular sensitivity. Fourier attenuation is one possible source of the initial weakness, not a substitute for this more general description.

A separate issue is the distance the parameters must travel. Reaching a fixed positive localization ratio  $\lambda = \gamma h$  as  $h \rightarrow 0$  requires  $\gamma$  to grow like  $h^{-1}$ . Small remaining residuals and bounded readouts limit how quickly ordinary raw-parameter GD can accumulate that displacement. Those drift bounds are part of the theory, not an optional numerical diagnostic.

The note separates **exact identities**, **conditional bounds with proofs**, and **hypotheses about actual training paths**. The Fourier sensitivity literature is cited where it applies; the explicit bounds below are derived here for the stated models. No result below says that every tanh network, optimizer, or target must encounter the barrier. An instantaneous gradient bound becomes a training-time obstruction only when its assumptions persist along the trajectory.

### 1.1 Model, residual, and optimization coordinates

The baseline is the raw shallow network

$$f(x) = d + \sum_{j=1}^W c_j \tanh(a_j x + b_j), \quad \theta = (a, b), \quad v = (d, c). \quad (1.1)$$

On fixed samples  $x_i$ , with  $|x_i| \leq X$ , define

$$e(x) = f(x) - g(x), \quad r_i = \frac{e(x_i)}{\sqrt{m}}, \quad \mathcal{R} = \|r\|_2, \quad \mathcal{L} = \frac{1}{2} \|r\|_2^2. \quad (1.2)$$

Thus  $e$  is the *spatial residual*,  $r$  is the normalized sample-residual vector, and  $\mathcal{R}$  is training root-mean-square error. The output weights are  $c_j$ , not the input slopes  $a_j$ . Write

$$\gamma_j = |a_j|, \quad z_j = -b_j/a_j, \quad \lambda_j = h\gamma_j$$

for nonzero slopes, using one fixed reference spacing  $h$  throughout a run. Larger  $\gamma$  means narrower transitions; the sum of those transitions can still represent a smooth function. A capital  $\Gamma$  below denotes a *desired inverse-width threshold*, not the current value of  $\gamma$ .

Unless otherwise stated, gradient flow has unit learning-rate factors. For discrete GD,  $\ell_k$  denotes the geometry learning rate, and accumulated flow time is replaced by  $\sum_k \ell_k$ . The empirical drift bounds use raw  $(a, b)$  coordinates. The Fourier and population-scale calculations use the center-preserving derivative of  $c \tanh(\gamma(x - z))$ . These are different parameter metrics, identified explicitly rather than conflated.

## 2 Fourier sensitivity: an explicit statement about the residual

### 2.1 The correct reference and the relevant claim

The primary tanh-specific optimization reference is **Xu, Zhang, Luo, Xiao, and Ma (2020), Section 6, especially equations (26)–(31) and Theorem 1 [1]**. Their frequency-principle analysis writes modal loss gradients in terms of the Fourier residual and identifies an exponential factor  $\exp[-\pi|\omega|/(2|a_j|)]$  in the small-weight regime. The general-network treatment of Luo et al. [2] addresses activation regularity and sampling densities under its own assumptions. The project workshop appendix proves the  $\text{sech}^2$  transform used below [4, Appendix A.1.3].

The statement needed here is stronger than “the kernel transform has an exponential tail”: *a high-frequency residual must have an exponentially small pairing with the current scale tangent*. We now derive that pairing and its bound. This absolute residual-band bound is our deduction; it is not a quotation of Xu et al.’s relative low-versus-high-frequency theorem.

### 2.2 From the scale derivative to the Fourier pairing

Temporarily use the whole-line loss  $\mathcal{L}_{\mathbb{R}} = \frac{1}{2} \int_{\mathbb{R}} e(x)^2 dx$ , assuming this is finite and differentiation under the integral is valid. For one neuron, put

$$\psi_{\gamma,z}(x) = (x - z) \text{sech}^2(\gamma(x - z)).$$

Holding its center and readout fixed when differentiating gives

$$\partial_{\gamma} \mathcal{L}_{\mathbb{R}} = c \int_{\mathbb{R}} e(x) \psi_{\gamma,z}(x) dx. \quad (2.1)$$

Use the angular-frequency convention  $\widehat{u}(\omega) = \int u(x) e^{-i\omega x} dx$  and set

$$F(\xi) = \frac{\pi \xi}{\sinh(\pi \xi / 2)}, \quad F(0) = 2.$$

The transform of  $\text{sech}^2(\gamma x)$  is  $\gamma^{-1} F(\omega/\gamma)$ . Multiplication by  $x$  differentiates its transform, so

$$\widehat{\psi}_{\gamma,z}(\omega) = \frac{i}{\gamma^2} e^{-i\omega z} F'(\omega/\gamma). \quad (2.2)$$

Consequently,

$$\partial_{\gamma} \mathcal{L}_{\mathbb{R}} = \frac{c}{2\pi} \int_{\mathbb{R}} \widehat{e}(\omega) \overline{\widehat{\psi}_{\gamma,z}(\omega)} d\omega. \quad (2.3)$$

This is the missing residual statement: the learning signal depends on the *product of the residual spectrum and the scale-tangent spectrum*, including their phases.

For  $\xi \geq 1$ , put  $s = \pi \xi / 2$ . Direct differentiation yields

$$|F'(\xi)| = \pi \frac{s \coth s - 1}{\sinh s} \leq C_F \xi e^{-\pi \xi / 2}, \quad C_F := \frac{\pi^2 \coth(\pi/2)}{1 - e^{-\pi}}.$$

Indeed,  $s \coth s - 1 \leq s \coth(\pi/2)$  and  $1/\sinh s \leq 2e^{-s}/(1 - e^{-\pi})$ . By symmetry the same bound holds with  $|\xi|$ .

For a single Fourier mode, the modal squared error therefore satisfies

$$\left| \partial_{\gamma} \left( \frac{1}{2} |\widehat{e}(\omega)|^2 \right) \right| \leq \frac{C_F |c| |\widehat{e}(\omega)|}{\gamma^2} \frac{|\omega|}{\gamma} e^{-\pi |\omega|/(2\gamma)}, \quad |\omega| \geq \gamma. \quad (2.4)$$

This follows from  $\partial_{\gamma} \widehat{e} = c \widehat{\psi}_{\gamma,z}$  and  $|\text{Re}(\widehat{e} \overline{\partial_{\gamma} \widehat{e}})| \leq |\widehat{e}| |\partial_{\gamma} \widehat{e}|$ . In particular, fixing the residual amplitude does not remove the exponential frequency penalty. Integrating that observation gives a statement for an entire residual band.

**Proposition 1** (High-frequency residual gives a weak scale gradient). Assume  $e \in L^2(\mathbb{R})$ ,  $\hat{e} \in L^1(\mathbb{R})$ , and  $\hat{e}(\omega) = 0$  for  $|\omega| < \Omega$ , with  $\Omega \geq \gamma > 0$ . Define the residual Fourier mass

$$M_\Omega := \frac{1}{2\pi} \int_{|\omega| \geq \Omega} |\hat{e}(\omega)| d\omega.$$

Then

$$|\partial_\gamma \mathcal{L}_\mathbb{R}| \leq \frac{C_F |c| M_\Omega}{\gamma^2} \frac{\Omega}{\gamma} \exp\left(-\frac{\pi\Omega}{2\gamma}\right). \quad (2.5)$$

*Proof.* Apply absolute values in (2.3) and insert (2.2). The function  $q \mapsto qe^{-\pi q/2}$  is decreasing for  $q \geq 1$ . Its maximum on  $|\omega|/\gamma \geq \Omega/\gamma \geq 1$  is therefore attained at  $\Omega/\gamma$ . Pull that maximum out of the integral.  $\square$

A residual can have order-one amplitude while (2.5) is exponentially small. The obstruction is the ratio  $\Omega/\gamma$ , not the visual smoothness of the target. The bound does not predict the *sign* of scale drift: an exponentially weak signal can point upward, downward, or cancel exactly.

### 2.3 Low-frequency leakage, boundaries, and sampling are load-bearing

For a general residual, split  $e = e_{<\Omega} + e_{\geq\Omega}$  by Fourier restriction. The exact spatial integral

$$\int_{\mathbb{R}} |x - z| \operatorname{sech}^2(\gamma(x - z)) dx = \frac{2 \log 2}{\gamma^2}$$

gives

$$|\partial_\gamma \mathcal{L}_\mathbb{R}| \leq \frac{|c|}{\gamma^2} \left[ 2 \log 2 \|e_{<\Omega}\|_\infty + C_F M_\Omega \frac{\Omega}{\gamma} e^{-\pi\Omega/(2\gamma)} \right]. \quad (2.6)$$

Thus high-frequency dominance in a residual plot is not enough: even a small low-frequency component may dominate the gradient pairing. Likewise, with a sampling density  $p$ , the pairing contains  $p(x)e(x)$ ; it is the spectrum of this *weighted residual* that enters the same calculation.

There is no automatic exponential theorem for a finite window. For illustration, take a harmonic residual  $e(x) = B_0 \cos(\omega x + \varphi)$  on an interval  $I$ , with center  $z \in I$  at distance at least  $d_\partial$  from its endpoints. For the unnormalized integral loss, subtracting the two omitted whole-line tails gives

$$\left| c \int_I e(x) \psi_{\gamma,z}(x) dx \right| \leq |c| B_0 \left[ \frac{|F'(\omega/\gamma)|}{\gamma^2} + \left( \frac{4d_\partial}{\gamma} + \frac{2}{\gamma^2} \right) e^{-2\gamma d_\partial} \right]. \quad (2.7)$$

To prove the tail term, use  $\operatorname{sech}^2(\gamma t) \leq 4e^{-2\gamma|t|}$  and integrate  $|t|$  over both tails  $|t| \geq d_\partial$ . A normalized uniform loss divides (2.7) by  $|I|$ ; a sampled loss additionally needs a quadrature or sampling-error bound. Tapering changes the spectral diagnostic and does not remove these conditions from the original objective.

### 2.4 An exponential escape-time consequence, under a persistent spectral gap

An instantaneous small gradient becomes an escape-time bound only with trajectory assumptions. Suppose a centered-scale gradient flow starts at  $\gamma_0 > 0$ , remains positive, and until its first hit of  $\Gamma \leq \Omega$  the *entire effective residual* satisfies Proposition 1, with  $M_\Omega(t) \leq M$  and  $|c(t)| \leq C_{\text{out}}$ . Then

$$\dot{\gamma} \leq C_F C_{\text{out}} M \Omega \gamma^{-3} e^{-\pi\Omega/(2\gamma)}.$$

Let  $T_\Gamma$  be the first hitting time. Integrating a strictly increasing potential whose derivative is the reciprocal of the right-hand side gives

$$T_\Gamma \geq \frac{1}{C_F C_{\text{out}} M \Omega} \int_{\gamma_0}^\Gamma s^3 e^{\pi\Omega/(2s)} ds. \quad (2.8)$$

This does not require monotone scale motion. If  $\Gamma \geq 2\gamma_0$ , retaining only  $s \in [\gamma_0, 2\gamma_0]$  yields the explicit weaker bound

$$T_\Gamma \geq \frac{\gamma_0^4}{C_F C_{\text{out}} M \Omega} \exp\left(\frac{\pi \Omega}{4\gamma_0}\right). \quad (2.9)$$

This is exponential in frequency relative to the initial scale, up to the displayed algebraic factors. **It is not a universal exponential lower bound in network width.** If low-frequency residuals, boundary terms, or other optimization coordinates supply substantial scale motion, this particular bound does not apply. The empirical task is to determine whether its persistent-gap premise is a useful approximation to an actual stalled trajectory.

### 3 Readout projection: which geometry signal is removed?

#### 3.1 An exact decomposition before any readout solve

Return to the finite sample problem. Let  $A(\theta)$  be the  $m \times (W + 1)$  feature matrix divided by  $\sqrt{m}$ , with the constant feature included, and  $y_i = g(x_i)/\sqrt{m}$ . Then  $r = Av - y$ . Define

$$J = \partial_\theta(Av) \text{ with } v \text{ fixed}, \quad P = AA^\dagger, \quad Z = (I - P)J, \quad C = A^\dagger J. \quad (3.1)$$

$P$  is an orthogonal projector even if the feature columns are highly nonorthogonal. These matrices map parameters to *all sample predictions*; scalar input and output do not make them rank one.

Since  $J = AC + Z$  and  $A^T Z = 0$ , ordinary gradients satisfy

$$\boxed{\nabla_\theta \mathcal{L} = C^T \nabla_v \mathcal{L} + Z^T r_\perp, \quad r_\perp = (I - P)r.} \quad (3.2)$$

The first term is geometry force shared with readout fitting. The second comes from residual outside the readout span, through prediction changes the readout cannot reproduce. This is a decomposition, not a claim that either term always increases scale.

Under joint unit-rate gradient flow,

$$\dot{r} = -(AA^T + JJ^T)r. \quad (3.3)$$

At temporarily fixed geometry, readout-only flow has the exact solution

$$r(t) = r_\perp + e^{-AA^T t} P r(0). \quad (3.4)$$

Hence expressible residual modes are removed at rates set by squared feature singular values. Ordinary readout GD need not solve every difficult readout direction to remove the *particular* modes that supplied useful early scale motion.

This motivates, but does not prove, the freezing hypothesis: if the first term in (3.2) drives useful scale growth early, rapid readout adaptation can remove it before substantial growth occurs. Freezing early can preserve that opportunity; freezing after residual reduction cannot recreate it. Whether the preserved force has the right sign is part of the hypothesis.

#### 3.2 Exact variable projection and its assumptions

Define  $v_*(\theta) = A(\theta)^\dagger y$ ,  $r_* = Av_* - y$ , and

$$\Phi(\theta) = \frac{1}{2} \|r_*\|_2^2.$$

Work locally at constant rank, with the smooth minimum-norm readout. Since  $A^T r_* = 0$ , differentiation gives

$$\boxed{\nabla_\theta \Phi = J_*^T r_* = Z_*^T r_*}. \quad (3.5)$$

This is the variable-projection gradient identity; see O’Leary and Rust [3]. The star on  $J_*$  matters: changing the readout also changes the geometry Jacobian, whose columns contain  $c_j$ .

For completeness, the full reduced-residual derivative in coordinate  $\theta_k$  is

$$\partial_{\theta_k} r_* = (I - P)(\partial_{\theta_k} A)v_* - (A^\dagger)^T (\partial_{\theta_k} A)^T r_*. \quad (3.6)$$

The first term follows by projecting  $dr_* = dA v_* + A dv_*$  outside the feature span. Differentiating  $A^T r_* = 0$  determines its component inside the span and gives the second term. The latter is orthogonal to  $r_*$ , so it contributes nothing to the scalar-loss gradient, but it can affect a Gauss–Newton approximation. At zero residual it vanishes. This distinction is standard in the full versus Kaufman approximate VarPro Jacobian [3].

An in-training LS intervention also changes residuals and coefficients simultaneously. At fixed geometry,

$$J_+^T r_+ - J_-^T r_- = J_-^T (r_+ - r_-) + (J_+ - J_-)^T r_+. \quad (3.7)$$

The first term isolates residual replacement; the second isolates the altered readout-scaled tangent map. Large fitted readouts can make the second term important. It is therefore unjustified to attribute every observed post-solve change solely to residual magnitude.

A truncated or regularized readout fit is a different objective. A projector onto a retained subspace is a useful diagnostic, but the exact identities for  $v_*$  must not be silently applied to a rank-changing solver with a different projector.

### 3.3 Magnitude, compensation, and signed alignment

For a chosen scale direction  $d$  in geometry coordinates, set

$$S_d = \|J_* d\|, \quad \chi_d = \frac{\|Z_* d\|}{\|J_* d\|}, \quad \mathcal{A}_d = -\frac{\langle r_*, Z_* d \rangle}{\|r_*\| \|Z_* d\|}.$$

These quantities are defined when their denominators are nonzero. Here  $\mathcal{A}_d$  is a signed alignment. Then

$$\boxed{-D\Phi[d] = \mathcal{R}_* S_d \chi_d \mathcal{A}_d.} \quad (3.8)$$

$S_d$  is raw sensitivity,  $\chi_d$  the non-compensable fraction, and  $\mathcal{A}_d$  whether the remaining residual favors the proposed scale increase. Small residual magnitude is only one of four factors. At a profiled state, freezing the readout does not instantaneously change this gradient; it changes the subsequent coupled dynamics.

## 4 Sensitivity squaring: the general mechanism, not a Fourier assumption

### 4.1 Why forward and backward sensitivities multiply

Suppose  $\theta_*$  is an exactly fitted profiled state, the rank remains locally constant, and write  $e_\theta = \theta - \theta_*$ . By (3.6),

$$r_*(\theta) = Z_* e_\theta + O(\|e_\theta\|^2), \quad \Phi(\theta) = \frac{1}{2} \|Z_* e_\theta\|^2 + O(\|e_\theta\|^3).$$

Thus the local linearized dynamics is

$$\nabla\Phi \simeq Z_*^T Z_* e_\theta, \quad \dot{e}_\theta = -Z_*^T Z_* e_\theta. \quad (4.1)$$

If  $Z_* v_k = s_k u_k$ , then  $e_\theta = e_k v_k$  first creates residual  $s_k e_k u_k$ , and backpropagation creates gradient  $s_k^2 e_k v_k$ . Therefore

$$\boxed{\dot{e}_k = -s_k^2 e_k, \quad e_{k,n+1} = (1 - \ell s_k^2) e_{k,n}.} \quad (4.2)$$

On the identifiable subspace,  $\kappa_2(Z_*^T Z_*) = \kappa_2(Z_*)^2$ . This slowdown occurs in exact arithmetic; it is not caused by explicitly forming normal equations. It also does not apply indiscriminately to every weak mode: the error must load onto that mode. Away from zero residual, additional nonlinear curvature terms matter.

For a single scale tangent  $j_\gamma$ , write

$$\|(I - P)j_\gamma\| = \|j_\gamma\| \sin \vartheta_\gamma.$$

The corresponding one-direction curvature at an exact fit is

$$\|(I - P)j_\gamma\|^2 = \|j_\gamma\|^2 \sin^2 \vartheta_\gamma. \quad (4.3)$$

Small readout magnitude, weak raw scale response, and near-readout redundancy can therefore compound multiplicatively.

A scalar residual norm does not by itself measure this difficulty. Under profiled unit-rate gradient flow, while  $\mathcal{R}_* > 0$ ,

$$\boxed{-\frac{d}{dt} \log \mathcal{R}_* = \|Z_*^T \hat{r}_*\|^2, \quad \hat{r}_* = r_*/\mathcal{R}_*} \quad (4.4)$$

This follows from  $\dot{\Phi} = -\|\nabla \Phi\|^2$  and  $\Phi = \mathcal{R}_*^2/2$ . If the right-hand sensitivity is at most  $\kappa$  throughout a region, reducing  $\mathcal{R}_*$  from  $R_0$  to  $\varepsilon$  while staying there requires

$$T \geq \kappa^{-2} \log(R_0/\varepsilon).$$

A small gradient due only to a small residual need not imply slow *relative* convergence; a small residual-normalized sensitivity does.

## 4.2 A low-frequency example and the meaning of “residual squared”

Let  $u_0, u_1$  be orthonormal smooth low-frequency functions and consider

$$f_{v,q} = (v + q)u_0 + \delta q u_1, \quad y = y_0 u_0 + \delta q_* u_1, \quad 0 < \delta \ll 1.$$

The readout optimum is  $v_*(q) = y_0 - q$ . Consequently,

$$r_* = \delta(q - q_*)u_1, \quad Z = \delta u_1, \quad \Phi'(q) = \delta^2(q - q_*). \quad (4.5)$$

The large change along  $u_0$  is readout-compensable; only the small change along  $u_1$  teaches  $q$ . **No high-frequency residual is involved.** This toy model isolates a possible mechanism, not a proved description of a particular tanh trajectory.

In general  $Z^T r_*$  is linear in  $r_*$  with  $Z$  fixed. If instead an unresolved component has amplitude  $\delta$  in *both* the residual and its parameter response,  $r_*(q) = \delta u(q)$ , then

$$\nabla \Phi = \delta^2 D u(q)^T u(q).$$

That is the precise quadratic-amplitude effect. It is not a universal law  $\|\nabla \Phi\| \propto \mathcal{R}_*^2$ .

In a translation-invariant linearized model, Fourier modes can be the singular modes. A tangent multiplier  $m_\gamma(\omega)$  then gives

$$\partial_t \hat{r}(\omega) = -|m_\gamma(\omega)|^2 \hat{r}(\omega).$$

The Fourier view identifies a reason that  $m_\gamma(\omega)$  is small; the sensitivity view explains why GD pays its square. For a finite network, singular modes need not be Fourier modes, and readout compensation can be severe in low-frequency directions too.

### 4.3 The small-scale tanh expansion exhibits readout redundancy directly

For one centered feature on a bounded domain, write  $u = x - z$  and assume  $|\gamma u| \ll 1$ . Taylor expansion gives

$$\partial_\gamma \tanh(\gamma u) = \frac{1}{\gamma} \tanh(\gamma u) - \frac{2}{3} \gamma^2 u^3 + O(\gamma^4 u^5). \quad (4.6)$$

The leading term lies in the current readout span. Therefore the projected scale column is

$$Z_{\gamma,j} = -\frac{2}{3} c_j \gamma_j^2 (I - P) u^3 + O(|c_j| \gamma_j^4)$$

for bounded  $u$ , in the sampled norm. More compensation by other features may shrink it further. The activation derivative can be close to one while the *independent scale signal* is already weak. This is a readout–scale redundancy mechanism, not saturated-activation vanishing.

## 5 Bounds on how far $\gamma$ can drift during training

### 5.1 A general residual-budget bound for raw-parameter GD

These are the main polynomial/time-budget statements from our earlier derivations. They require no Fourier assumption and permit joint movement of every center and readout.

**Proposition 2** (Raw slope travel is limited by accumulated residual). For the empirical loss (1.2), raw geometry gradient flow obeys

$$|a_j(T) - a_j(0)| \leq X \int_0^T |c_j(t)| \mathcal{R}(t) dt. \quad (5.1)$$

For ordinary simultaneous GD, the corresponding exact bound is

$$|a_{j,K} - a_{j,0}| \leq X \sum_{k < K} \ell_k |c_{j,k}| \mathcal{R}_k. \quad (5.2)$$

*Proof.* The slope update uses

$$\partial_{a_j} \mathcal{L} = \frac{c_j}{m} \sum_i e(x_i) x_i \operatorname{sech}^2(a_j x_i + b_j).$$

Cauchy–Schwarz,  $|x_i| \leq X$ , and  $\operatorname{sech}^2 \leq 1$  give  $|\partial_{a_j} \mathcal{L}| \leq X |c_j| \mathcal{R}$ . Integrating the gradient-flow equation or summing discrete updates proves the claims. The inequality  $||a| - |b|| \leq |a - b|$  gives the same displacement bound for  $\gamma_j = |a_j|$ .  $\square$

Assume  $|c_j(t)| \leq C_{\text{out}}$  and loss-decreasing training, so  $\mathcal{R}(t) \leq R_0$ . Then

$$\gamma_j(T) \leq \gamma_j(0) + X C_{\text{out}} R_0 T, \quad \boxed{T_\Gamma \geq \frac{(\Gamma - \gamma_j(0))_+}{X C_{\text{out}} R_0}}. \quad (5.3)$$

With  $\Gamma = \lambda_*/h$ ,  $\gamma_j(0) = O(1)$ , and width-independent  $X, C_{\text{out}}, R_0$ , this yields the necessary time  $T_\Gamma = \Omega(h^{-1})$ . Here the  $\Omega(\cdot)$  symbol denotes a lower-bound order, not the frequency cutoff used in Section 2.

A stronger scaling occurs if the actual frozen readouts obey  $|c_j| \leq C_0 h$ :

$$|\lambda_j(T) - \lambda_j(0)| \leq X C_0 h^2 R_0 T, \quad \boxed{T_{\lambda_*} \geq \frac{(\lambda_* - \lambda_j(0))_+}{X C_0 h^2 R_0}}. \quad (5.4)$$

Thus a fixed positive increase in  $\lambda$  costs at least order  $h^{-2}$  under this raw-coordinate metric. This readout scaling arises for the bounded-density quadrature witness

$$q'(x) = h \sum_j \zeta(z_j) \frac{\gamma}{2} \operatorname{sech}^2(\gamma(x - z_j)), \quad c_j = \frac{h}{2} \zeta(z_j),$$

when  $\|\zeta\|_\infty$  is bounded independently of width. The derivative-to-tanh construction is standard in the project [4]. **The bounded-density condition must be checked for the coefficients actually used;** it is not automatically true for every cardinal filter or LS-refitted geometry.

These are necessary-time lower bounds, not predictions that observed learning time equals  $W$  or  $W^2$ . They concern reaching a specified scale, not the impossibility of fitting the target by another representation. Converting  $h^{-1}$  to order  $W$  also assumes the center/halo convention gives  $h^{-1} = \Theta(W)$ .

## 5.2 Late freezing and residual depletion

If the readout is frozen at time  $t_f$ , geometry-only flow decreases the loss. Proposition 2 then gives, for continuation time  $H$ ,

$$|\lambda_j(t_f + H) - \lambda_j(t_f)| \leq hX|c_j(t_f)|H\mathcal{R}(t_f). \quad (5.5)$$

Late freezing at small residual therefore leaves little subsequent scale travel at fixed readout magnitude and training budget. Early freezing removes this particular restriction, but it does not supply a lower bound on useful upward movement: that still requires positive alignment in (3.8).

More generally, if a trajectory satisfies

$$\mathcal{R}(t) \leq R_0 e^{-\mu t} + R_{\text{floor}}, \quad |c_j(t)| \leq C_{\text{out}},$$

then

$$|\gamma_j(T) - \gamma_j(0)| \leq XC_{\text{out}} \left[ \frac{R_0}{\mu} (1 - e^{-\mu T}) + TR_{\text{floor}} \right]. \quad (5.6)$$

If  $R_{\text{floor}} = 0$ , the available displacement is finite. Any threshold beyond  $\gamma_j(0) + XC_{\text{out}}R_0/\mu$  is unreachable under these assumptions. This is a conditional non-reachability result, not a proof that the residual-decay hypothesis holds in a network that still needs greater scale.

## 5.3 Centered population flow: cubic and quartic time bounds

There is also a stronger overlap calculation for the independently optimized centered scale. For the population loss

$$\mathcal{L}_p = \frac{1}{2} \int e(x)^2 p(x) dx,$$

assume  $p(x) \leq p_{\text{max}}$ ,  $|e(x, t)| \leq E_\infty$ , and  $|c_j(t)| \leq C_{\text{out}}$  uniformly along the interval of interest. Changing variables  $u = \gamma(x - z)$  in (2.1) gives

$$|\dot{\gamma}| = |\partial_\gamma \mathcal{L}_p| \leq \frac{2 \log 2 C_{\text{out}} E_\infty p_{\text{max}}}{\gamma^2}. \quad (5.7)$$

Multiplying by  $3\gamma^2$  and integrating yields

$$T_\Gamma \geq \frac{\Gamma^3 - \gamma_0^3}{6 \log 2 C_{\text{out}} E_\infty p_{\text{max}}} \quad (\Gamma > \gamma_0). \quad (5.8)$$

No fixed sample gap is needed. The constants must hold along the trajectory; the readout and other centered parameters may otherwise move.

An additional cancellation is available when  $F_t(x) := p(x)e(x, t)$  is globally Lipschitz on  $\mathbb{R}$  with constant  $L_{\text{res}}(t)$ . Since  $u \text{sech}^2 u$  is odd,

$$\partial_\gamma \mathcal{L}_p = \frac{c}{\gamma^2} \int u \text{sech}^2 u [F_t(z + u/\gamma) - F_t(z)] du.$$

The expansion  $F(\xi) = 2 - \pi^2 \xi^2/12 + O(\xi^4)$  implies  $\int_{\mathbb{R}} u^2 \text{sech}^2 u du = -F''(0) = \pi^2/6$ . Thus

$$|\partial_\gamma \mathcal{L}_p| \leq \frac{\pi^2 |c| L_{\text{res}}(t)}{6\gamma^3}, \quad \gamma(T)^4 - \gamma_0^4 \leq \frac{2\pi^2 C_{\text{out}}}{3} \int_0^T L_{\text{res}}(t) dt. \quad (5.9)$$

If  $L_{\text{res}}(t) \leq L_0$ , then  $T_\Gamma \geq 3(\Gamma^4 - \gamma_0^4)/(2\pi^2 C_{\text{out}} L_0)$ .

$L_{\text{res}}$  **is not the target Lipschitz constant.** For smooth  $p$ ,

$$L_{\text{res}}(t) \leq \|p\|_\infty [\text{Lip}(f_t) + \text{Lip}(g)] + \|p'\|_\infty \|e_t\|_\infty.$$

It can depend on width and training time. A hard finite-interval cutoff need not yield globally Lipschitz  $pe$ , so the quartic argument then requires boundary terms or a different assumption. These restrictions prevent an instantaneous bound from being over-interpreted as a uniform theorem about training.

## 5.4 A different exponential bound: narrow transitions inside a sample gap

For fixed center  $z$ , suppose every sample satisfies

$$d_{\text{gap}} \leq |x_i - z| \leq D, \quad d_{\text{gap}} > 0,$$

and  $|c(t)| \leq C_{\text{out}}$ ,  $m^{-1} \sum_i |e(x_i, t)| \leq E$ . Centered-scale flow then satisfies

$$|\dot{\gamma}| \leq 4C_{\text{out}} D E e^{-2d_{\text{gap}}\gamma},$$

and differentiating  $e^{2d_{\text{gap}}\gamma(t)}$  gives

$$T_\Gamma \geq \frac{e^{2d_{\text{gap}}\Gamma} - e^{2d_{\text{gap}}\gamma_0}}{8d_{\text{gap}} C_{\text{out}} D E}. \quad (5.10)$$

This concerns *already narrow transitions missed by all samples*, not broad features failing to sense high-frequency error. A persistent gap is essential; center motion can destroy it. For midpoint samples of spacing  $h/s$ ,  $2\gamma d_{\text{gap}} = \lambda/s$ , so the exponent need not be large in the useful QI regime. It is a separate diagnostic, not the default explanation of the small- $\gamma$  hold-up.

## 5.5 Which drift result applies?

| Bound       | Dynamics and required assumptions                                                                          | What it says                                                                               |
|-------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| (5.3)–(5.4) | Raw empirical GD; bounded readouts and residual                                                            | At least order $h^{-1}$ time, or $h^{-2}$ for $O(h)$ readouts, to reach $\lambda = O(1)$ . |
| (5.8)–(5.9) | Centered population flow; bounded residual envelope, and Lipschitz weighted residual for the quartic bound | Scale grows no faster than $t^{1/3}$ or $t^{1/4}$ under uniform constants.                 |
| (2.8)       | Centered flow; persistent high-frequency residual and bounded Fourier mass                                 | Escape from broad initial scales can be exponential in $\Omega/\gamma_0$ .                 |
| (5.10)      | Centered empirical flow; persistent sample gap                                                             | Reaching increasingly narrow scales can take exponential time in $d_{\text{gap}}\Gamma$ .  |

These are not competing estimates of one universal rate. They describe different information, coordinate, and trajectory regimes.

## 6 Whitening and the connection between the two explanations

### 6.1 Separate raw sensitivity from readout-compensation loss

Let  $G$  be a chosen scale-tangent matrix in a fixed coordinate convention, and  $Z_s = (I - P)G$ . A convenient diagnostic is the center-preserving log-scale tangent

$$G_{ij} = \frac{c_j}{\sqrt{m}}(a_j x_i + b_j) \operatorname{sech}^2(a_j x_i + b_j), \quad (6.1)$$

obtained from  $a_j, b_j \mapsto e^{\varepsilon_j}(a_j, b_j)$ . It is a probe definition, not a change to the baseline GD algorithm.

Take the thin SVD  $G = U_G \Sigma_G V_G^T$  over its nonzero singular directions. Then

$$\boxed{Z_s^T Z_s = V_G \Sigma_G \underbrace{(I - U_G^T P U_G)}_{C_{\text{comp}}} \Sigma_G V_G^T.} \quad (6.2)$$

$\Sigma_G$  measures raw scale sensitivity. The eigenvalues of  $C_{\text{comp}}$  are squared sines of principal angles to the readout span: they measure the *additional* loss caused by compensation. Neither small raw singular values nor small principal angles establish a bottleneck unless the actual residual excites the associated directions.

The Fourier hypothesis concerns residual-relevant response before projection, as in (2.5). The compensation hypothesis concerns the loss after projection. They can compound, and the projected singular modes need not remain Fourier modes. One should not assume that projection simply multiplies every Fourier response by a scalar.

### 6.2 What full and block whitening explain

For a fixed readout problem  $\frac{1}{2} \|Av - y\|^2$ , let  $A = U \Sigma V^T$  and set  $v = Tz$ ,  $T = V \Sigma^{-1}$  on the retained subspace. Then

$$AT = U, \quad \nabla_z^2 \mathcal{L} = I.$$

Full whitening removes the singular-value spread of this *sample-level* problem. Independent block whitening instead gives  $\tilde{A} = [Q_1, \dots, Q_B]$ , with  $Q_b^T Q_b = I$  but nonzero cross-block terms  $Q_b^T Q_c$ . For two blocks the Gram eigenvalues are  $1 \pm \sigma_k(Q_1^T Q_2)$ , with additional unit eigenvalues if block dimensions differ. Nearly duplicate block spans therefore leave arbitrarily weak global directions.

Whitening raw scale tangents  $G$  similarly does not eliminate compensation:

$$T_G = V_G \Sigma_G^{-1} \implies (Z_s T_G)^T (Z_s T_G) = C_{\text{comp}}.$$

This is why separating within-block sensitivity from cross-block coupling is useful. Global feature whitening succeeds regardless of what caused the original weak singular values; its success alone cannot identify Fourier attenuation as the cause of slow gamma drift.

The original handoff points to two motivating observations: D14's aggressive in-training readout solves harmed geometry-quality improvement in the tested scheme, while incomplete solves helped; D23's global whitening strongly outperformed block whitening for a fixed dictionary [6, 7]. D14 does not isolate all possible causes of the effect, and D23 trains no geometry. These observations motivate the analysis above rather than establish it.

### 6.3 Parameterization and online whitening are later interventions

For a positive raw slope,  $\gamma = a$  and  $z = -b/a$ . Thus raw slope motion generally mixes scale and center motion. Independently optimizing  $\rho = \log \gamma$  instead gives

$$\dot{\gamma} = -\gamma^2 \partial_{\gamma} \mathcal{L},$$

so the centered cubic/quartic time bounds change under that metric. Likewise,  $\nabla \mathcal{L}/\mathcal{R} = \nabla \mathcal{R}$  and  $\nabla \mathcal{L}/\mathcal{R}^2 = \nabla \log \mathcal{R}$  away from zero residual. A positive scalar rescaling of a geometry-only gradient flow changes its clock, not its paths; discrete steps and blockwise rescaling can change trajectories. This is why the coordinate convention belongs in every drift claim [5].

A training-only whitening transform can be exported into the readout:  $ATz = Av$  with  $v = Tz$ . Refreshing  $T$  requires coordinate transport to preserve predictions and special care if rank is truncated. We defer this optimizer design question until the dominant suppression mechanism is identified.

## 7 Working hypotheses and a short empirical handoff

**H-F: spectral suppression.** The remaining residual contains frequencies high relative to current scales, and its raw scale-gradient contribution is suppressed approximately as in (2.5), after accounting for the actual sampling/window. Low-frequency leakage in (2.6) must not explain most of the gradient instead.

**H-R: readout compensation and depletion.** Readout adaptation removes positive, useful scale-driving force, or the relevant scale directions have a small non-compensable fraction in (3.8). Earlier freezing should improve useful scale growth *when this directional premise holds*, not merely preserve a larger loss.

**H-D: insufficient drift budget.** The measured residual/readout trajectory leaves too little accumulated raw-slope movement to reach useful scales within the budget. Equations (5.1)–(5.6) quantify this without requiring high-frequency error.

**H-W: global cancellation.** The difficult fitting directions cross parameter/feature blocks. Block normalization leaves them weak, whereas global whitening resolves them. This is a conditioning mechanism to connect to  $Z_s$ , not evidence by itself of gamma learning.

### 7.1 Three experiments, with room for independent inference

**1. Freeze timing.** Compare early, intermediate, late, and no readout freezing from paired nonzero-readout initializations, with matched raw-geometry update budgets. Record actual evaluation error, signed scale motion, centers, readout magnitudes, and the residual budget. A readout refit on a saved copy distinguishes useful dictionary improvement from mere movement. Earlier freezing that increases scales but not dictionary quality weakens the proposed mechanism.

**2. Fourier versus projection.** At representative early and stalled checkpoints, compare the actual residual spectrum with raw and projected scale responses. Equal-RMS sine/cosine residual probes at fixed network state isolate frequency effects; inspect responses versus  $|\omega|/\gamma_j$ . Use (3.8) and (6.2) to determine whether suppression is already present before projection or is mainly added by compensation. Let the measured data, not the idealized theorem, determine which explanation is relevant.

**3. Whitening control.** On a fixed dictionary, compare raw, block-, and full-whitened readout fitting with stable, documented step sizes. Inspect achieved whitening, cross-block correlations, and residual loading on weak modes. This calibrates cancellation and numerical effects before attempting online geometry whitening; it is not itself a scale-discovery experiment.

Use an independent dense evaluation grid, report the loss/parameter normalization, and keep rank cutoffs explicit and consistent. Save states rather than only endpoint errors. The requested output is a short mechanistic interpretation supported by a few diagnostic plots, including mixed or negative results. No new optimizer, broad hyperparameter sweep, or claim of a universal barrier is required.

**What has and has not been proved.** The displayed identities and conditional bounds have derivations above. What remains empirical is whether real trajectories maintain their premises and whether the weak directions matter for the task. General convergence or non-convergence of joint GD, emergence of a particular gamma distribution, and optimal freezing schedules are not established.

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- [4] Continuous MLPs Project. *Constructing Machine-Precision Neural Networks with Quasi-Interpolants*. Workshop draft, 2026; supplied project file `workshop_continuous.pdf`. Appendix A.1.3, p. 15, proves the normalized  $\text{sech}^2$  transform; Section 3.2 explains integration into a tanh network. The optimization bounds in this handoff are separate derivations, not claims taken from the draft’s approximation theorem.
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- [6] `precision-mlps`, archived D14 iteration 1 results. `iteration_1_results.md`. Draft empirical evidence cited in the original handoff; not a theorem about all LS/GD schedules.
- [7] `precision-mlps`, [D23 whitening README](#) and `results/summary.csv`. Fixed-dictionary readout evidence cited in the original handoff; gamma is selected externally, not trained. No new repository runs were performed for this revision.